

Course Details

Course Overview:

Course Code:	GPGN 318 Applied Geophysics I
Semester Year:	Fall 2026
Credit Hours:	3
Class Meeting Times:	Lecture: Monday, Wednesday 3:00 pm-3:50 pm, Lab: Tuesdays 1:00 pm-3:50 pm
Class Location:	Lecture: Marquez 122, Lab: Marquez 310

Instructor: Roslynn King **Email:** roslynn.king@mines.edu

Office Location: Green Center 245A **Office Hours:** TBD

Pre-requisites:

- GPGN 229
- GPGN 268 Recommended
 - Python knowledge helpful
 - Experience with conda environments
 - numpy/scipy/matplotlib

Co-requisites:

- GPGN 328

Required Technology:

- Canvas (For lab handout distribution and submission of Lab Reports)
- Laptop (For running computer based labs)

Welcome to Applied Geophysics I

I look forward to working with you all throughout the semester. We will be covering three types of geophysical methods, all related to static (not time-varying) fields. This is why there is a corequisite of GPGN 328. Throughout the course we will be doing hands-on exercises with many different geophysical instruments. I encourage you to play around with the instruments yourself to gain a better understanding of how to work with them. Do not worry, as long as you are not physically aggressive with them, you will not break them.

Course Description:

From the catalog:

Applied Geophysics I is an introductory course on the application of static fields to image the Earth's subsurface. The static fields include electrostatics, magnetostatics, and gravitational fields. These tools are employed in various geotechnical and environmental engineering problems, resource exploration and production monitoring, geothermal site characterization, hazards, and humanitarian efforts. Through the combination of two-hours of lecture and a three-hour lab each week, the students are provided with the fundamental theory and hands-on field experiments for each of these techniques, including the principles, instrumentation, and procedures of data acquisition, analysis, and interpretation.

Brief list of topics covered:

- a. General Survey Design
 - i. How do we know where we are?
 - ii. How often do we collect data?
 - iii. How do we estimate noise?
 - iv. Relative vs. Absolute measurements
- b. The Gravity Method
 - i. Introduction to the theory of gravitational field
 - ii. The earth's gravitational field and its global variations
 - iii. Basic principles of gravimeters
 - iv. Survey design and field procedure for acquiring gravity data
 - v. Data reduction in gravity survey
 - vi. Determination of rock density
 - vii. Modeling of gravity anomalies
 - viii. Interpretation and inversion of gravity data
- c. The Electrical Method
 - i. Electric current in conductive media
 - ii. Vertical electrical sounding
 - iii. 1D inversion and interpretation
 - iv. 2D and 3D imaging
 - v. Inversion and interpretation of 2D data
 - vi. Spontaneous (self) potential (SP)
 - vii. Induced polarization (IP)
- d. The Magnetic Method
 - i. Introduction to the theory of magnetic field
 - ii. Magnetization and the field produced by magnetized objects
 - iii. Susceptibility of rocks and minerals, and magnetization of the material in earth's crust
 - iv. Magnetic anomalies in applied geophysics: anomalous field and its projections
 - v. Principles of magnetometers, survey design, and field procedures

- vi. Modeling of magnetic anomalies
- vii. Self-demagnetization and remnant magnetization
- viii. Interpretation and inversion of magnetic data

Learning Outcomes:

At the completion of the course, you will be able to:

- Apply knowledge of mathematics, science, and engineering (ABET outcome 1).
- Identify and solve applied science problems (ABET outcome 2).
- Work as a team to solve applied geoscience problems (ABET outcome 5).
- Design and conduct experiments, as well as to analyze and interpret data (ABET outcome 6).
- Use the techniques, skills, and modern scientific and technical tools necessary for professional practice (ABET outcome 7).

Assessments:

Homework/Lab Exercises: 50%

Attendance/Participation: 20%

Final Project: 20%

Group Quizzes: 10%

Laboratory Exercises:

The laboratory exercises are a necessary component of this course, with the goal of providing practical experience with the application of gravity, magnetic and electrical methods in geophysical problems.

The labs are divided into three fundamental categories common to many geophysical problems:

1. data acquisition and field procedure
2. data reduction and processing
3. numerical modeling and interpretation

A report must be submitted for every laboratory exercise. For computer processing based labs, the content of the report will be based on the questions asked in the lab worksheet. For field labs you will submit the relative pages of your lab notebook, and you will often have to submit plots of the data collected (with minimal processing).

Grading:

Portions of this course assessment follows a [*specifications grading approach*](#), an alternative grading practice that aims to provide students with clear guidelines and assessment rubrics for successful work submissions and, when falling short, effective feedback for how to prepare your work for resubmission to successfully meet the established criteria.

Course assessment will be based on **4** Categories:

- **Category 1: Group quizzes**
 - Purpose: to demonstrate an understanding and use of key principles covered in the course lecture material and work together.

- **Category 2: Lab reports and Assignments**
 - Purpose: to demonstrate your understanding of the data collection, processing, and interpretation components.
- **Category 3: Attendance and Participation**
 - Purpose: to demonstrate consistent engagement with the course by being present for lectures and labs and contributing actively to the hands-on, collaborative work.
- **Category 4: Final Project**
 - Purpose: to demonstrate your ability to integrate the methods learned across the course, apply them to a geophysical problem, and communicate your results.

Group Quizzes

- The group quizzes will take place in class, with no notes or computer access.
- The date of the quizzes will be announced ahead of time.
- You will be randomly assigned to a group.
- Will be multiple choice.
- Will be graded right then and there, if any submissions are incorrect you will immediately get another chance at answering.
- You will receive full marks for the question if you answer it correctly the first time. For every attempt afterwards you will receive partial credit based on your remaining chances. For example, if there are 4 potential answers to a question, if you get it correct on the 1st try, you will get 1 point, if you get it correct on the 2nd try, you will get .75 points, 3rd try will be 0.5 points, and at the very least you will get .25 points for the last try.

Lab Reports and Assignments

- Submitted Category 2 assignments will be assessed as one of four following “grades”:
 1. **Exceeds Expectations:** A high-quality submission that clearly meets all stated specifications with minimal-to-no need of any improvements. Will receive credit for a "Pass" grade.
 2. **Meets Expectations:** The submitted work meets all stated specifications, though could be improved as per the provided feedback. Will receive credit for a "Pass" grade.
 3. **Near Miss:** Does not meet all stated specifications and will not receive credit for that submission. See Assignment Resubmission Policy below.
 4. **Incomplete:** Meets few-to-no specifications and needs considerable improvement to be acceptable. Does not receive credit for that submission. See Assignment Resubmission Policy below.
- **Exceeds Expectations** assessments potentially can be used to improve final marks by a fractional letter grade (i.e., from B to B+ or A- to A).

Assignment Submission Policy

- **Due Dates:** Assignments should be submitted by their due dates for guaranteed full credit.
- **Requests for an extension** must be made to the instructor at **least 48 hours ahead of the scheduled due date** and will be entertained on a case-by-case basis.
- Unless an extension is granted in advance, assignments not submitted by the due date will receive an **Incomplete** grade.

Assignment Resubmission Policy

- Students will be provided assignment feedback that can be used for revision of their work.
- Resubmissions will be accepted within 5 business days of receiving feedback.
- Previously submitted work may only be resubmitted **once** for reevaluation.
- Students receiving a **Meets Expectations**, **Near Miss** or **Incomplete** on an assignment may resubmit that assignment using the provided instructor and teaching assistant feedback for an improved grade.
- Include a short statement on what you changed.
- Only one resubmission of an **Incomplete** assignment is permitted for the course.
- **There is no penalty for (acceptable) resubmission of assignments; full credit is possible.**

Grading Policy:

Your grade for each section will be determined by the number of “pass” assessments in that section:

I expect there to be about 14 Lab assignments

Grade	Number complete (# EE)
A	14 (6 EE)
A-	14 (4 EE)
B+	14 (2 EE)
B	14
C	11
D	7
F	<7

Participation:

- Attendance will be taken at the **end** of every lab period, and randomly at lectures. Active participation starts with being present for class. If you must miss a class or lab, let me know ahead of time.

Final Project:

- The final project is a joint project shared with GPGN 328 (Physics of the Earth), the theory course taught in lock step with this one. Rather than treating the applied and theoretical sides separately, the project asks you to combine them: you will apply the field methods, data processing, and interpretation skills from Applied Geophysics I together with the underlying theory from Physics of the Earth to address a single geophysical problem. Details on the project topic, deliverables, and timeline will be provided later in the semester. You will receive an Applied Geophysics 1 grade and a separate Physics of the Earth grade for the Final Project.

Coursework Return Policy:

Grading may take time to complete, and an effort will be made to return work to students within two weeks at most. When Lab notebooks are collected, they will be returned before the next lab period.

Required Material:

You will be required to have a fresh notebook for taking notes during labs. I particularly recommend a rite-in-the-rain notebook. Small rite-in-the-rain notebooks can be purchased from the bookstore, but larger ones are available through other sellers.

There are no required textbooks for this course. All informational material will be provided for you.

Recommended Resources:

These are resources you might want to borrow or purchase. You should at least become familiar with these and the key ideas presented in each.

- Burger, H.R, Sheehan, A.F., and Jones, C.H., 2006, *Introduction to Applied Geophysics: Exploring the Shallow Subsurface*, W.W. Norton

Additional Suggested Resources:

A few resources that have useful insights into our learning in this course include:

- <https://gpg.geosci.xyz>

Course Schedule

This is a *predicted* schedule for the course, it will most likely change as we go through the semester.

Week	Monday lecture	Tuesday lab (green if outside)	Wednesday lecture
1 (Aug 24)	01 Intro to Geophysics	Lab 1 — GPS Adventure (outside)	(no class)
2 (Aug 31)	02 Fund 1 — GPS & surveying	Lab 2 — Surveying (outside)	03 Fund 2 — Data & statistics
3 (Sep 7)	Labor Day (no class)	Lab 3 — Repeatability (outside)	04 Fund 3 — Noise & survey design
4 (Sep 14)	05 Gravity 1 — Basics & instrumentation	Career Day (no class)	06 Gravity 2 — Surveying
5 (Sep 21)	07 Gravity 3 — Procedures	Lab 4 — Radius of the Earth (outside)	08 Gravity 4 — Design & corrections
6 (Sep 28)	09 Gravity 5 — More corrections	Lab 5 — Gravity modeling	10 Gravity 6 — Interpretation
7 (Oct 5)	11 Gravity 7 — More gravity methods	Lab 6 — Gravity surveying (tunnel) (outside)	Gravity Quiz + catch-up
8 (Oct 12)	Fall Break (no class)	Fall Break (no class)	12 DC Fundamentals

9 (Oct 19)	13 DC Experiment	Lab 7 — Gravity interpretation	14 DC Survey Design
10 (Oct 26)	15 DC Surveying	Lab 8 — Sounding Profiling (outside)	16 DC Interpretation
11 (Nov 2)	17 Case Study	Lab 9 — Sounding Interpretations	18 IP
12 (Nov 9)	DC Quiz + Final Project launch	Lab 10 — 2D DC collection (outside)	19 Magnetics
13 (Nov 16)	20 Magnetics Source	Lab 11 — 2D DC Inversion	21 Magnetics Survey
14 (Nov 23)	22 Magnetics Interpretation	Lab 12 — Magnetics Anomalies	<i>Thanksgiving</i>
15 (Nov 30)	Magnetics Quiz + project work	Lab 13 — Magnetics Surveying (outside)	Project work / special topics
16 (Dec 7)	Final Project presentations	Lab 14 — Magnetics Inversion	<i>(term ends Dec 8 – no class)</i>

Policies and Campus Resources

Absences:

The first commitment to students at Mines is to their academics. Class attendance is required of all students unless the student has an excused absence granted by the school or the student's professor. An excused absence awarded by the school or professor comes after a student's request or initiative. To review the Excused Absence Policy and/or to request an excused absence, please visit <https://www.mines.edu/student-life/student-absences/>.

Sexual Misconduct, Discrimination, and Retaliation:

Discrimination, Harassment, and Sexual Misconduct of any type, including sexual harassment, sexual assault, dating violence, domestic violence, and stalking, are prohibited under the Policy Prohibiting Unlawful Discrimination and the Policy Prohibiting Sexual Misconduct, Discrimination, and Retaliation. Please see the [Office for Institutional Equity website](#) for information on Sexual Misconduct and Discrimination.

As your instructor, it is my responsibility to foster a learning environment that supports diversity of thoughts, perspectives and experiences, honors your identities, and is free of discrimination. Please feel free to speak to me if the class is not living up to these standards. Please note that I am also a mandatory reporter as an instructor at Mines, and if I receive a disclosure of Discrimination, Harassment, and/or Sexual Misconduct, I am required to report it to the Title IX Coordinator.

Preferred First Name Project:

Mines recognizes members of the campus community may prefer to use a first name other than their legal name to identify themselves. Many services on campus, like Canvas, utilize and display preferred first names. Additional information on preferred name, including how to update your preferred name, is available at the [Office For Institutional Equity website](#).

Academic Integrity:

Colorado School of Mines affirms the principle that all individuals associated with the Mines academic community have a responsibility for establishing, maintaining, and fostering an understanding and appreciation for academic integrity. In broad terms, this implies protecting the environment of mutual trust within which scholarly exchange occurs, supporting the ability of the faculty to fairly and effectively evaluate every student's academic achievements, and giving credence to the university's educational mission, its scholarly objectives, and the substance of the degrees it awards. We desire an environment free of any and all forms of academic misconduct and expect students to act with integrity at all times. Please read the [full academic misconduct/integrity policy](#) for full definitions of academic misconduct. Additionally, please use [resources provided by the Office of Community Standards](#) for guidance should you need to know more about the procedures of the policy for academic misconduct/integrity.

Generative Artificial Intelligence:

The Office of the Provost encourages the entire University community to explore the uses and impacts of GenAI technologies, whether through critical discussions or creative applications. Based on a review of the most recent [Guidelines for Using Generative Artificial Intelligence at Colorado School of Mines](#), the GenAI policy for this course will be:

Generative Artificial Intelligence (genAI) tools such as ChatGPT are important resources in many fields and industries. Because these tools will be used in professional and personal contexts, we believe it is valuable for you to engage critically with these tools and explore their use in generating content submitted for evaluation in this course. You remain responsible for all content you submit for evaluation.

You may use genAI tools to help generate ideas and brainstorm. However, you should note that the material generated by these tools may be inaccurate, incomplete, biased, or otherwise problematic. We encourage you to consider how genAI complements, supplants, or fails to replace your contributions and abilities.

If you include content (e.g., ideas, text, code, images) that was generated, in whole or in part, by Generative Artificial Intelligence tools (including, but not limited to, ChatGPT and other large language models) in work submitted for evaluation in this course, [you must document and credit your source](#). Failure to properly cite sources, including AI tools for generating content, would be considered Academic Misconduct in violation of [Mines Academic Integrity/Misconduct Policy](#).

Course Issues and Concerns:

As part of good professional practice, students are encouraged to speak with the faculty directly to raise issues and concerns with regards to the course professionally in compliance with the student code of conduct. Students can also reach out to the head of the department, Paul Sava, the course is being offered through. The department head can investigate and work with the faculty member to resolve course-related concerns. Students' final point of contact is the college dean, Terri Hogue, who can make any final decisions.

Disability Support Services:

Disability Support Services (DSS) works collaboratively with students, faculty, and staff to minimize barriers and support an accessible campus community. When barriers to access occur, Disability Support Services works one-on-one with students to determine accommodations and facilitate access to programs and services. If you've been approved for accommodations through Disability Support Services, please contact your professor to confirm receipt of your accommodation letter and to discuss the implementation of accommodations in this course. Please visit mines.edu/disability-support-services for more information or to request accommodations.

Digital Accessibility:

The Colorado School of Mines is committed to supporting an accessible digital environment for all members of our community, including students with disabilities. If you have an accessibility concern with Canvas or any digital materials or software used in this course, please contact your professor or request support from Information & Technology Solutions. Please visit <https://www.mines.edu/accessibility/> for more information.

Student Outreach & Support (SOS) Resources:

If you feel overwhelmed, anxious, depressed, distressed, mentally or physically unhealthy, or concerned about your wellbeing overall, there are resources both on- and off-campus available to you. If you need assistance, please ask for help from a trusted faculty or staff member, fellow student, or submit a referral for yours. As a community of care, we can help one another get through difficult times. If you are concerned for another student, offer assistance and/or ask for help on their behalf. Students seeking resources for themselves or others should visit mines.edu/sos.

Student Outreach and Support can help connect you with a variety of resources; some of those might include:

- Counseling Center – <https://www.mines.edu/counseling-center/> or students may call to make an appointment. There are also online resources for students on the website. Located in the Wellness Center 2nd floor. Located at 1770 Elm St.
- Health Center - <https://www.mines.edu/student-health/> or students may call to make an appointment. Located in Wellness Center 1st floor.
- Colorado Crisis Services - For crisis support 24 hrs/7 days, either by phone, text, or in person, Colorado Crisis Services is a great confidential resource, available to anyone. <http://coloradocrisisservices.org>, 1-844-493-8255, or text "TALK" to 38255. Walk-in location addresses are posted on the website.

In an emergency, you should call 911, and they will dispatch a Mines or Golden PD officer to assist.

Diversity and Inclusion:

At Colorado School of Mines, we understand that a diverse and inclusive learning environment inspires creativity and innovation, which are essential to the engineering process. We also know that in order to address current and emerging national and global challenges, it is important to learn with and from people who have different backgrounds, thoughts, and experiences.

Our students represent every state in the nation and more than 90 countries around the world, and we continue to make progress in the areas of diversity and inclusion by providing [Diversity and Inclusion programs and services](#) to support these efforts.

The Writing Center:

The Writing Center is a free academic support service for undergraduate and graduate students. Professional consultants and peer tutors provide support with all forms of communication including technical and scientific reports, academic essays, and oral presentations. Students can make an online or in-person appointment at any stage of their project, from brainstorming to final revisions.

To learn more about their services and to make an appointment, please visit writing.mines.edu. For questions, please e-mail writing@mines.edu

Participating in this Course

Expectations for Participation:

You are expected to engage in all course activities, tasks, and assignments as an emerging professional. You are expected to spend between 2-3 hours on this course each week during out-of-class time, in addition to actively participating during the class-time each week.

Profile in Canvas:

As part of the learning experience at the Colorado School of Mines, our class will be utilizing online learning resources and experiences through the Canvas learning management system. In order to help build community in this online learning environment, you are encouraged to upload your profile picture to Canvas. Photos should be similar to the photos taken for passports or state identification cards.

Expectations of online etiquette or netiquette:

Here are few do's and don'ts about communicating in your course through emails or in online discussion forums:

- Do...
 - Ask questions and engage in conversations as often as possible—feel free to contact the instructor via the discussion forum for questions or via email or other communication.
 - Be patient and respectful of others and their ideas and opinions they post online.
 - Remember to be thoughtful and use professional language. Keep in mind that things often come across differently in written text, so review your writing before posting.
 - Be prepared for some delays in response time, as "virtual" communication tends to be slower than "face-to-face" communication.
 - Contact the instructor if you feel that inappropriate content or behavior has occurred as part of the course.
 - Check the syllabus and course policies stated by your instructor to know what to expect about your instructor's turnaround time for responding.

- Do NOT...
 - Use inappropriate language—this includes, but is not limited to, the use of curse words, swearing, or language that is derogatory.
 - Post inappropriate materials—for example, accidentally posting/showing a picture that is not appropriate for the course content.
 - Post in ALL CAPS, as this is perceived as shouting and avoid abbreviations and informal language ("I'll C U L8R").
 - Send heated messages even if you are provoked. Likewise, if you should happen to receive a heated message, do not respond to it.
 - Send an email or post to the entire class, unless you feel that everyone must read it.